

COMPTE RENDU TECHNIQUE

Mise en place d'un Serveur LAMP

Debian 12 Trixie • Apache 2.4.65 • vsftpd 3.0.5 • CrowdSec 1.4.6

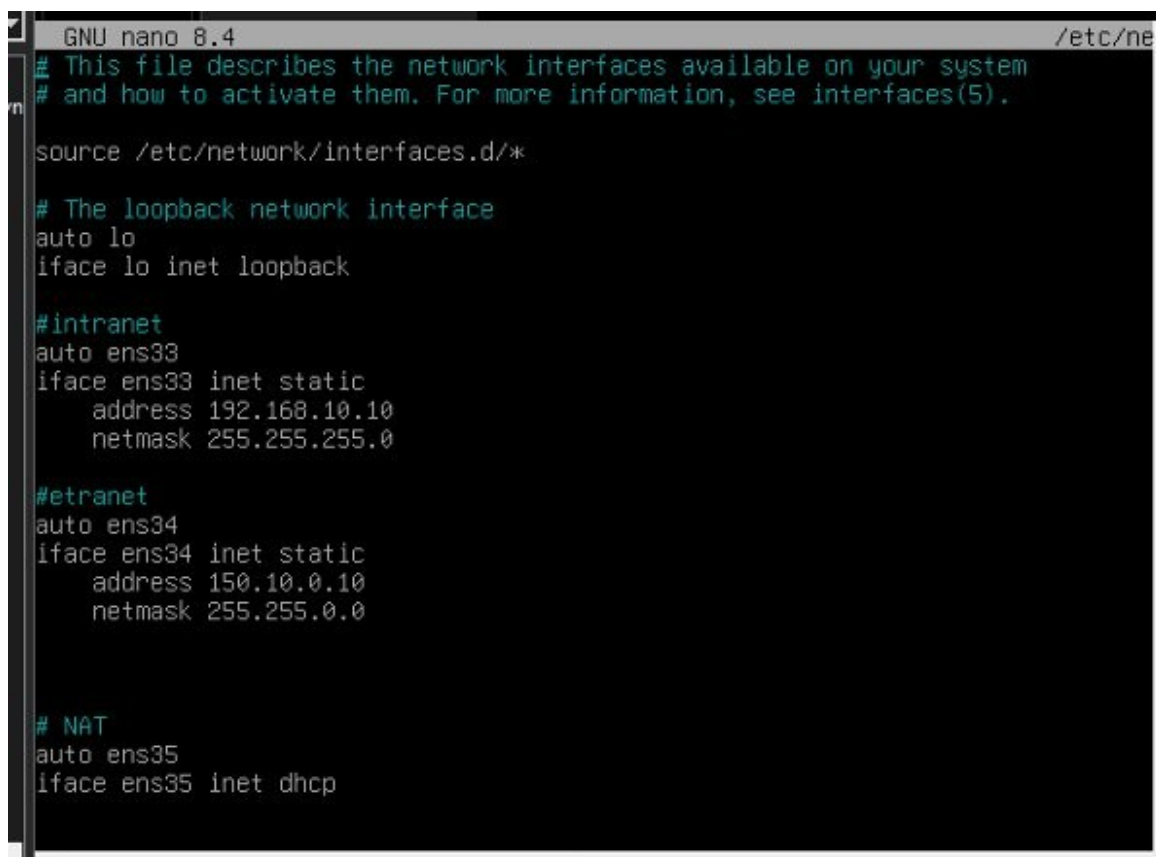
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1. Architecture Réseau

Le serveur dispose de trois interfaces réseau configurées dans `/etc/network/interfaces`, assurant l'isolation entre les différents périmètres.

Interface	Réseau	Adresse IP	Masque	Mode
ens33	Intranet	192.168.10.10	255.255.255.0	Statique
ens34	Extranet	150.10.0.10	255.255.0.0	Statique
ens35	NAT	DHCP	–	Dynamique

Configuration `/etc/network/interfaces`

A screenshot of a terminal window showing the configuration of network interfaces in the file `/etc/network/interfaces` using the GNU nano 8.4 editor. The file content is as follows:

```
GNU nano 8.4 /etc/ne
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

#intranet
auto ens33
iface ens33 inet static
    address 192.168.10.10
    netmask 255.255.255.0

#extranet
auto ens34
iface ens34 inet static
    address 150.10.0.10
    netmask 255.255.0.0

# NAT
auto ens35
iface ens35 inet dhcp
```

Fig. 1 – Configuration des interfaces réseau

Activation des interfaces et vérification

```

    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 00:0c:29:57:ef:47 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c2957ef47
3: ens34: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 00:0c:29:57:ef:51 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    altname enx000c2957ef51
4: ens35: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:57:ef:5b brd ff:ff:ff:ff:ff:ff
    altname enp2s3
    altname enx000c2957ef5b
    inet 192.168.22.132/24 brd 192.168.22.255 scope global dynamic noprefixroute ens35
        valid_lft 1192sec preferred_lft 967sec
    inet6 fe80::c96c:29ff:fe57:ef5b/64 scope link
        valid_lft forever preferred_lft forever
root@lamp:~# ip link set ens33 up
root@lamp:~# ip link set ens34 up
root@lamp:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:57:ef:47 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c2957ef47
    inet6 fe80::c96c:29ff:fe47:ef47/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:57:ef:51 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    altname enx000c2957ef51
    inet6 fe80::c96c:29ff:fe51:ef51/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
4: ens35: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:57:ef:5b brd ff:ff:ff:ff:ff:ff
    altname enp2s3
    altname enx000c2957ef5b
    inet 192.168.22.132/24 brd 192.168.22.255 scope global dynamic noprefixroute ens35
        valid_lft 1135sec preferred_lft 910sec
    inet6 fe80::c96c:29ff:fe5b:ef5b/64 scope link
        valid_lft forever preferred_lft forever
root@lamp:~# systemctl restart networking_

```

Fig. 2 – Activation des interfaces et résultat de `ip a`

Test de connectivité depuis Windows

```

C:\Users\karab>ping 192.168.10.10

Envoi d'une requête 'Ping' 192.168.10.10 avec 32 octets de données :
Réponse de 192.168.10.10 : octets=32 temps=2 ms TTL=64
Réponse de 192.168.10.10 : octets=32 temps=2 ms TTL=64
Réponse de 192.168.10.10 : octets=32 temps=2 ms TTL=64
Réponse de 192.168.10.10 : octets=32 temps=3 ms TTL=64

Statistiques Ping pour 192.168.10.10:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
    Durée approximative des boucles en millisecondes :
        Minimum = 2ms, Maximum = 3ms, Moyenne = 2ms

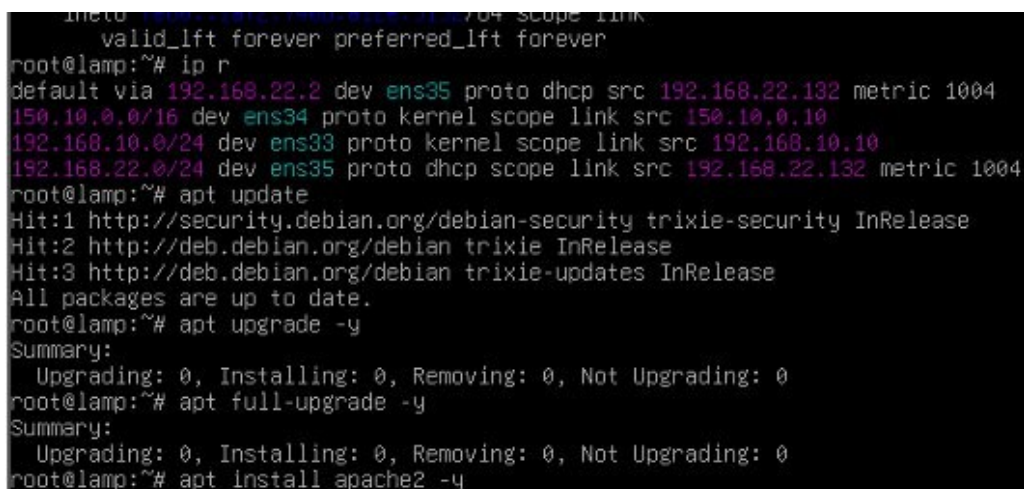
```

Fig. 3 – Ping 192.168.10.10 depuis Windows (0% perte, ~2ms)

2. Installation Apache2

Mise à jour du système, installation d'Apache2 et création des répertoires web.

```
apt update && apt upgrade -y && apt full-upgrade -y
apt install apache2 -y
mkdir -p /var/www/intranet && mkdir -p /var/www/extranet
chown -R www-data:www-data /var/www/ && chmod -R 755 /var/www/
```

A terminal window showing the execution of network configuration and package management commands. The output includes IP address details for ens33 and ens35, and the results of apt update, apt upgrade, and apt full-upgrade, all indicating that packages are up to date.

```
ineto 192.168.22.2 dev ens35 proto dhcp scope link
valid_lft forever preferred_lft forever
root@lamp:~# ip r
default via 192.168.22.2 dev ens35 proto dhcp src 192.168.22.132 metric 1004
150.10.0.0/16 dev ens34 proto kernel scope link src 150.10.0.10
192.168.10.0/24 dev ens33 proto kernel scope link src 192.168.10.10
192.168.22.0/24 dev ens35 proto dhcp scope link src 192.168.22.132 metric 1004
root@lamp:~# apt update
Hit:1 http://security.debian.org/debian-security trixie-security InRelease
Hit:2 http://deb.debian.org/debian trixie InRelease
Hit:3 http://deb.debian.org/debian trixie-updates InRelease
All packages are up to date.
root@lamp:~# apt upgrade -y
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
root@lamp:~# apt full-upgrade -y
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
root@lamp:~# apt install apache2 -y
```

Fig. 4 – Installation d'Apache2 et vérification des routes

3. VirtualHosts Apache

Deux VirtualHosts sont créés : l'intranet sur 192.168.10.10 et l'extranet sur 150.10.0.10, chacun avec son propre DocumentRoot.

intranet:



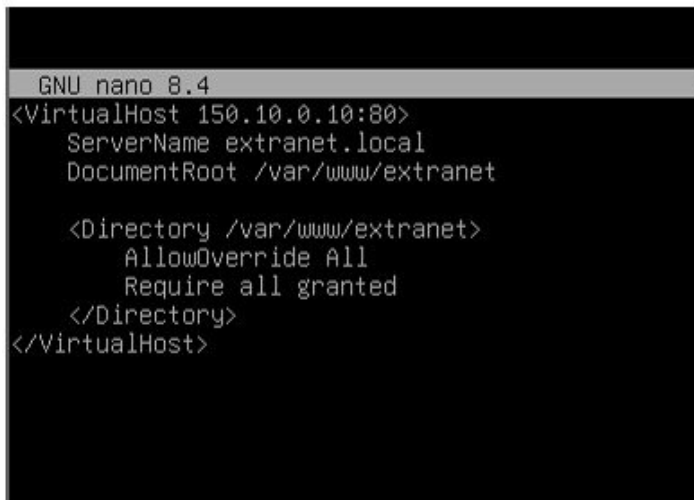
```

GNU nano 8.4 /etc/apache2/sites-available/intranet.conf *
<VirtualHost 192.168.10.10:80>
    ServerName intranet.local
    DocumentRoot /var/www/intranet

    <Directory /var/www/intranet>
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>_

```

extranet:



```

GNU nano 8.4
<VirtualHost 150.10.0.10:80>
    ServerName extranet.local
    DocumentRoot /var/www/extranet

    <Directory /var/www/extranet>
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>

```

Fig. 5 – Configurations VirtualHost intranet.conf et extranet.conf

Vérification avec curl



```

root@lamp:~# curl 192.168.10.10
<h1> INTRANET/h1p
sp:192.168.10.10:80/
root@lamp:~# curl 150.10.0.10
<h1> EXTRANET/h1p
sp:150.10.0.10:80/
root@lamp:~# openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout /etc/ssl/private/wildcard.key -out /etc/ssl/certs/wildcard.crt -subj "/C=FR/ST=IDF/L=Paris/O=Valserac/CN=*.valserac.local"
.....
.....
req warning: Skipping unknown subject name attribute "c"
root@lamp:~#

```

Fig. 6 – Test curl : contenu INTRANET et EXTRANET retournés correctement

4. HTTPS & Certificat SSL Wildcard

Génération d'un certificat wildcard auto-signé RSA 2048 bits valable 365 jours, couvrant *.valserac.local.

```

openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
    -keyout /etc/ssl/private/wildcard.key \
    -out /etc/ssl/certs/wildcard.crt \
    -subj "/C=FR/ST=IDF/L=Paris/O=Valserac/CN=*.valserac.local"

```

```

-----
req warning: Skipping unknown subject name attribute "c"
root@lamp:~# ls -l /etc/ssl/private/wildcard.key
-rw----- 1 root root 1704 Dec  1 13:54 /etc/ssl/private/wildcard.key
root@lamp:~# ls -l /etc/ssl/private/wildcard.crt
ls: cannot access '/etc/ssl/private/wildcard.crt': No such file or directory
root@lamp:~# ls -l /etc/ssl/private/wildcard.crt
ls: cannot access '/etc/ssl/private/wildcard.crt': No such file or directory
root@lamp:~# find / -name wildcard.crt 2>/dev/null
/etc/ssl/certs/wildcard.crt
root@lamp:~# ls -l /etc/ssl/certs/wildcard.crt
-rw-rw-r-- 1 root root 1265 Dec  1 13:54 /etc/ssl/certs/wildcard.crt
root@lamp:~# _

```

Fig. 7 – Vérification des fichiers clé privée et certificat

Modules Apache activés

```

a2enmod ssl && a2enmod rewrite && a2enmod headers
systemctl restart apache2

```

Ports déclarés dans ports.conf : 80, 443, 192.168.10.10:5501, 192.168.10.10:5502.

VirtualHost Intranet HTTPS (5501 → 5502)

```

GNU nano 8.4 /etc/apache2/sites-available/intranet.conf *
#http intra 5501 vers https 5502

<VirtualHost 192.168.10.10:5501>
    ServerName intranet.valserac.local

    RewriteEngine on
    RewriteRule ^ https://intranet.valserac.local:5502%{REQUEST_URI} [L,R=302]
</VirtualHost>

#https intra sur 5502
<VirtualHost 192.168.10.10:5502>
    ServerName intranet.valserac.local
    DocumentRoot /var/www/intranet

    <Directory /var/www/intranet>
        AllowOverride All
        Require all granted
    </Directory>

    SSLEngine on
    SSLCertificateFile /etc/ssl/certs/wildcard.crt
    SSLCertificateKeyFile /etc/ssl/private/wildcard.key

    ErrorLog ${APACHE_LOG_DIR}/intranet_error.log
    CustomLog ${APACHE_LOG_DIR}/intranet_access.log combined_
</VirtualHost>

```

Fig. 8 – intranet.conf : redirection HTTP 5501 → HTTPS 5502 + SSL

Résultats des tests HTTPS

Commande	Résultat attendu
curl -k http://150.10.0.10	302 → https://extranet.valserac.local/
curl -k https://150.10.0.10	Contenu EXTRANET
curl -k http://192.168.10.10:5501	302 → https://intranet.valserac.local:5502/
curl -k https://192.168.10.10:5502	Contenu INTRANET

5. Serveur FTP – vsftpd

Installation

```

root@lamp:~# apt install vsftpd -y
Installing:
  vsftpd

Installing dependencies:
  liburap0

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 199 kB
  Space needed: 465 kB / 8,259 MB available

Get:1 http://deb.debian.org/debian trixie/main amd64 liburap0 amd64 7.6.q-36 [55.3 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 vsftpd amd64 3.0.5-0.2 [144 kB]
Fetched 199 kB in 0s (2,281 kB/s)
Preconfiguring packages ...
Selecting previously unselected package liburap0:amd64.
(Reading database ... 30480 files and directories currently installed.)
Preparing to unpack .../liburap0_7.6.q-36_amd64.deb ...
Unpacking liburap0:amd64 (7.6.q-36) ...
Selecting previously unselected package vsftpd.
Preparing to unpack .../vsftpd_3.0.5-0.2_amd64.deb ...
Unpacking vsftpd (3.0.5-0.2) ...
Setting up liburap0:amd64 (7.6.q-36) ...
Setting up vsftpd (3.0.5-0.2) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/vsftpd.service' → '/usr/lib/systemd/system/vsftpd.service'.
/usr/lib/tmpfiles.d/vsftpd.conf:1: Line references path below legacy directory /var/run/, updating /var/run/vsftpd/empty →
tmpfiles.d/ drop-in file accordingly.
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12) ...
root@lamp:~# systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable vsftpd
root@lamp:~# systemctl start vsftpd
root@lamp:~# cp /etc/vsftpd.conf /etc/vsftpd.conf.orig
root@lamp:~# _

```

Fig. 9 – Installation et démarrage de vsftpd

Comptes utilisateurs et permissions

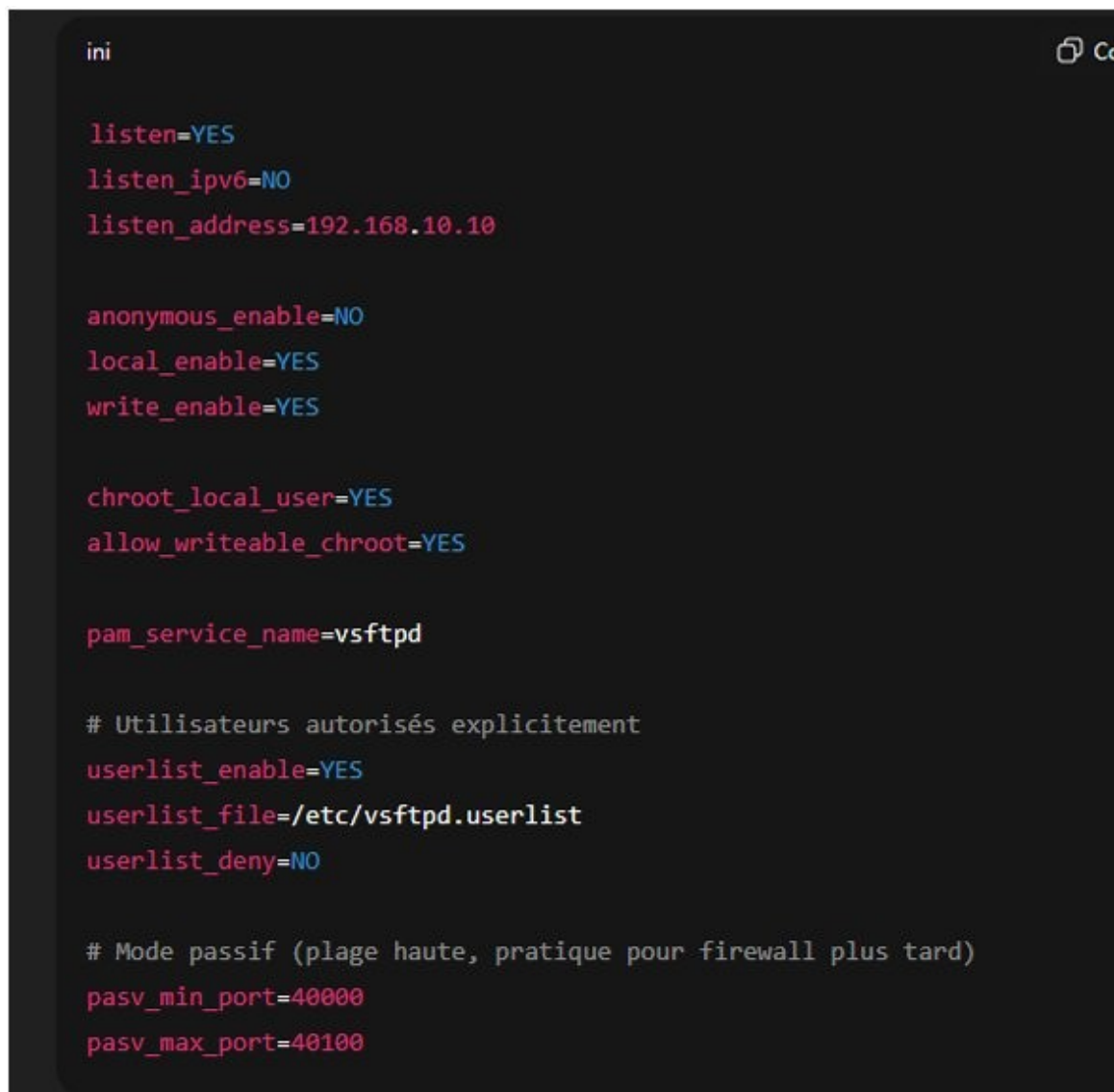
Utilisateur	Répertoire	Rôle
devweb	/var/www	Accès complet sources web
graphiste	/var/www/extranet/img	Accès limité aux images

```

useradd -d /var/www -s /usr/sbin/nologin devweb && passwd devweb
mkdir -p /var/www/extranet/img
useradd -d /var/www/extranet/img -s /usr/sbin/nologin graphiste
chown -R devweb:www-data /var/www
chown -R graphiste:graphiste /var/www/extranet/img
chmod -R 755 /var/www && chmod -R 775 /var/www/extranet/img

```

Configuration /etc/vsftpd.conf



```
ini

listen=YES
listen_ipv6=NO
listen_address=192.168.10.10

anonymous_enable=NO
local_enable=YES
write_enable=YES

chroot_local_user=YES
allow_writeable_chroot=YES

pam_service_name=vsftpd

# Utilisateurs autorisés explicitement
userlist_enable=YES
userlist_file=/etc/vsftpd.userlist
userlist_deny=NO

# Mode passif (plage haute, pratique pour firewall plus tard)
pasv_min_port=40000
pasv_max_port=40100
```

Fig. 10 – Configuration vsftpd : chroot, userlist, mode passif

6. Firewall – UFW

Politique par défaut : deny incoming. Ouvertures minimales selon les besoins.

```

root@lamp:~# ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
root@lamp:~# ufw allow 22
Rules updated
Rules updated (v6)
root@lamp:~# ufw allow 80
Rules updated
Rules updated (v6)
root@lamp:~# ufw allow 443
Rules updated
Rules updated (v6)
root@lamp:~# ufw allow from 192.168.10.0/24 to any port 5501 proto tcp
Rules updated
root@lamp:~# ufw allow from 192.168.10.0/24 to any port 5502 proto tcp
Rules updated
root@lamp:~# ufw allow from 192.168.10.0/24 to any port 21 proto tcp
Rules updated
root@lamp:~# ufw allow from 192.168.10.0/24 to any port 40000:40100 proto tcp
Rules updated
root@lamp:~# ufw enable
Firewall is active and enabled on system startup
root@lamp:~# ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22 ALLOW IN Anywhere
80 ALLOW IN Anywhere
443 ALLOW IN Anywhere
5501/tcp ALLOW IN 192.168.10.0/24
5502/tcp ALLOW IN 192.168.10.0/24
21/tcp ALLOW IN 192.168.10.0/24
40000:40100/tcp ALLOW IN 192.168.10.0/24
22 (v6) ALLOW IN Anywhere (v6)
80 (v6) ALLOW IN Anywhere (v6)
443 (v6) ALLOW IN Anywhere (v6)

root@lamp:~#

```

Fig. 11 – Règles UFW actives

Port / Plage	Action	Source
22	ALLOW IN	Anywhere
80	ALLOW IN	Anywhere
443	ALLOW IN	Anywhere
5501/tcp	ALLOW IN	192.168.10.0/24
5502/tcp	ALLOW IN	192.168.10.0/24
21/tcp	ALLOW IN	192.168.10.0/24
40000:40100/tcp	ALLOW IN	192.168.10.0/24

7. Protection Anti-DDoS – mod_evasive

mod_evasive bloque automatiquement les IPs qui dépassent les seuils de requêtes configurés.

```

apt install libapache2-mod-evasive -y
mkdir -p /var/log/mod_evasive
chown -R www-data:www-data /var/log/mod_evasive

```



```
GNU nano 8.4 /etc/apache2/mods-available/evasive.conf *
<IfModule mod_evasive.c>
  DDoSTableSize 3079
  DDoSPageCount 5
  DDoSSiteCount 1
  DDoSPageInterval 10
  DDoSSiteInterval 1
  DDoSBlockingPeriod 10

  DDoSEmailNotify root@localhost
  DDoSLogDir "/var/log/mod_evasive"
</IfModule>
```

Fig. 12 – Configuration *evasive.conf* (seuils DDoS)

8. Tests de Performance – ApacheBench

Test de charge : 200 requêtes avec 20 connexions simultanées sur le serveur intranet.

```
ab -n 200 -c 20 http://192.168.10.10/
```

```

 98%    28
 99%    30
100%    31 (longest request)
root@lamp:~# ab -n 200 -c 20 http://192.168.10.10/
This is ApacheBench, Version 2.3 <$Revision: 1923142 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking 192.168.10.10 (be patient)
Completed 100 requests
Completed 200 requests
Finished 200 requests


Server Software:      Apache/2.4.65
Server Hostname:      192.168.10.10
Server Port:          80

Document Path:        /
Document Length:      10703 bytes

Concurrency Level:    20
Time taken for tests:  0.110 seconds
Complete requests:    200
Failed requests:       0
Total transferred:    2195400 bytes
HTML transferred:     2140600 bytes
Requests per second:  1825.47 [#/sec] (mean)
Time per request:     10.956 [ms] (mean)
Time per request:     0.548 [ms] (mean, across all concurrent requests)
Transfer rate:        19568.51 [Kbytes/sec] received

Connection Times (ms)
              min    mean[+/-sd] median   max
Connect:        0      0   0.4      0      3
Processing:      1     10   9.7      6     41
Waiting:         0      7   7.9      5     40
Total:          1     10   9.7      6     41

Percentage of the requests served within a certain time (ms)
 50%    6
 66%   10
 75%   16
 80%   19
 90%   26
 95%   32
 98%   37
 99%   38
100%   41 (longest request)
root@lamp:~# _

```

Fig. 13 – Résultats ApacheBench : 1825 req/s, 0 échec

Métrique	Valeur
Requêtes totales	200
Requêtes échouées	0
Requêtes/seconde	1825.47 req/s
Temps moyen/requête	10.956 ms
Débit	19568.51 KB/s

Requête la plus longue	41 ms
------------------------	-------

9. Détection d'Intrusion – CrowdSec

CrowdSec complète UFW et mod_evasive en analysant les logs Apache pour détecter les comportements malveillants et bloquer les IPs via des listes communautaires.

```
apt install crowdsec
```

```

root@lamp:~# apt install crowdsec
Installing:
  crowdsec

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 38.1 MB
  Space needed: 137 MB / 8,243 MB available

Get:1 http://deb.debian.org/debian trixie/main amd64 crowdsec amd64 1.4.6-10+b4 [38.1 MB]
Fetched 38.1 MB in 1s (59.6 MB/s)
Selecting previously unselected package crowdsec.
(Reading database ... 31285 files and directories currently installed.)
Preparing to unpack .../crowdsec_1.4.6-10+b4_amd64.deb ...
Unpacking crowdsec (1.4.6-10+b4) ...
Setting up crowdsec (1.4.6-10+b4) ...
I: Registering to LAPI (/etc/crowdsec/local_api_credentials.yaml)
I: Registering to CAPI (/etc/crowdsec/online_api_credentials.yaml)
I: Setting up offline hub (see README.Debian)
I: Enabling upstream-recommended items, first installation (via symlinks from /etc/crowdsec)
I: Enabling WAL for SQLite [fstype=ext4] (see README.Debian)
Created symlink '/etc/systemd/system/multi-user.target.wants/crowdsec.service' → '/usr/lib/systemd/system/crowdsec.service'
W: Restarting manually to force a CAPI pull (upstream #2120)
root@lamp:~#

```

Fig. 14 – Installation CrowdSec avec enregistrement CAPI

- Enregistrement LAPI local : /etc/crowdsec/local_api_credentials.yaml
- Enregistrement CAPI cloud : /etc/crowdsec/online_api_credentials.yaml
- Hub offline et WAL SQLite activés
- Service systemd crowdsec.service activé au démarrage

10. Récapitulatif des Services

Service	Version	Statut	Démarrage auto
apache2	2.4.65	Actif	Oui
vsftpd	3.0.5	Actif	Oui
ufw	–	Actif	Oui
crowdsec	1.4.6	Actif	Oui

Réalisé sur Debian 12 Trixie